HOCHSCHULE
HAMM-LIPPSTADT

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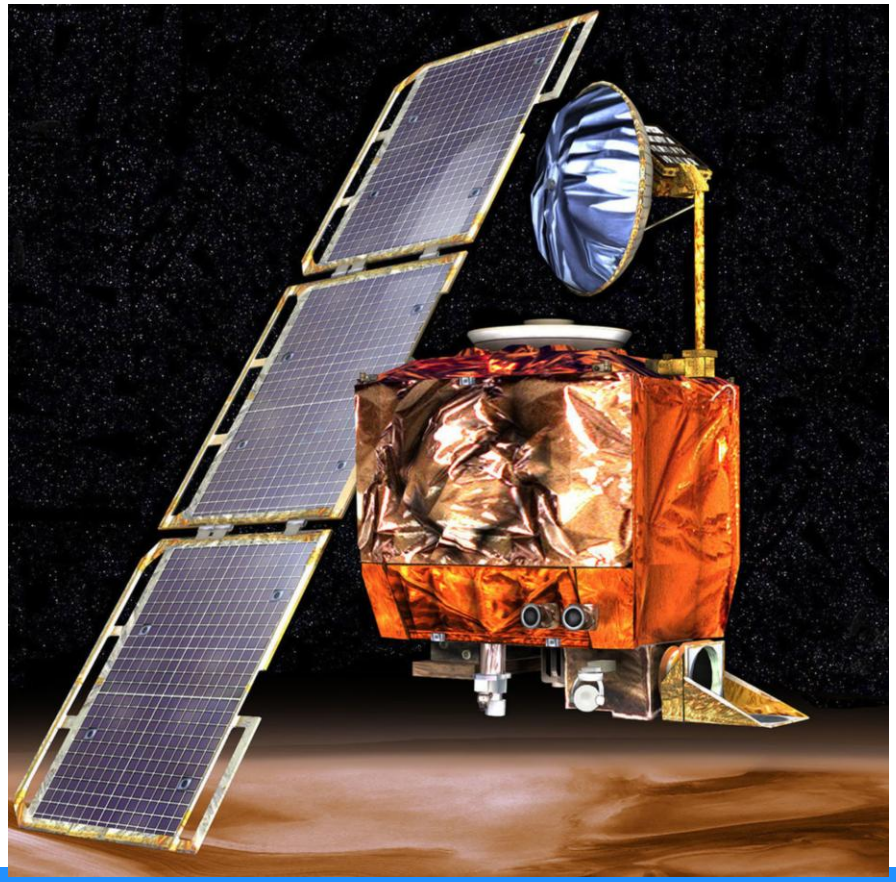
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HOCHSCHULE HAMM-LIPPSTADT

1. Introduction
2. Methods
3. **Analysis**
4. Design
5. Advanced Design Concepts
6. Discussion & Summary
7. Bibliography

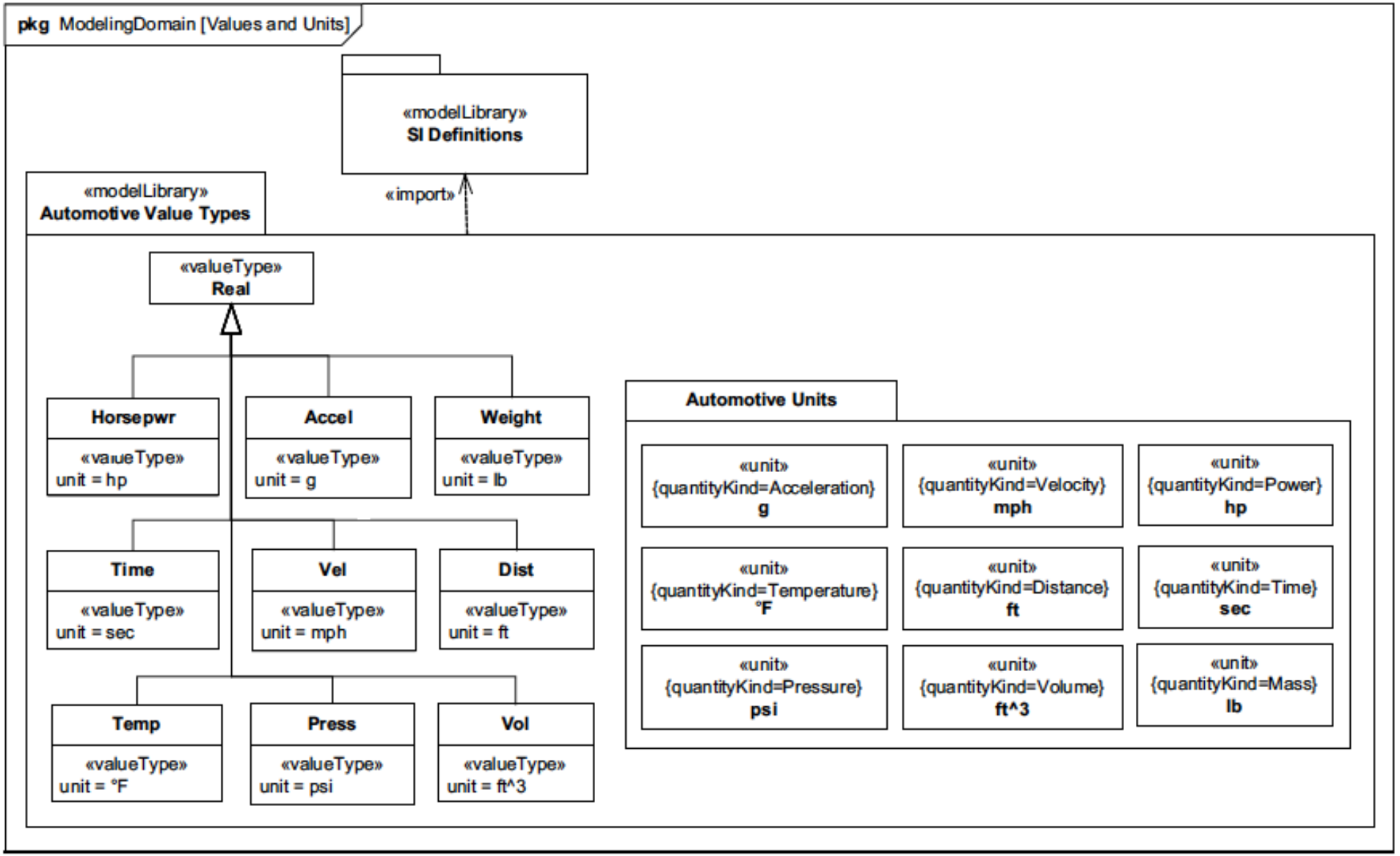
Mars Climate Orbiter Project (Failed)

- Launched 1998, cost $327 million
- goal: orbit Mars
- Destroyed 1999 — flew too close to Mars
- Cause: Different units of measurement



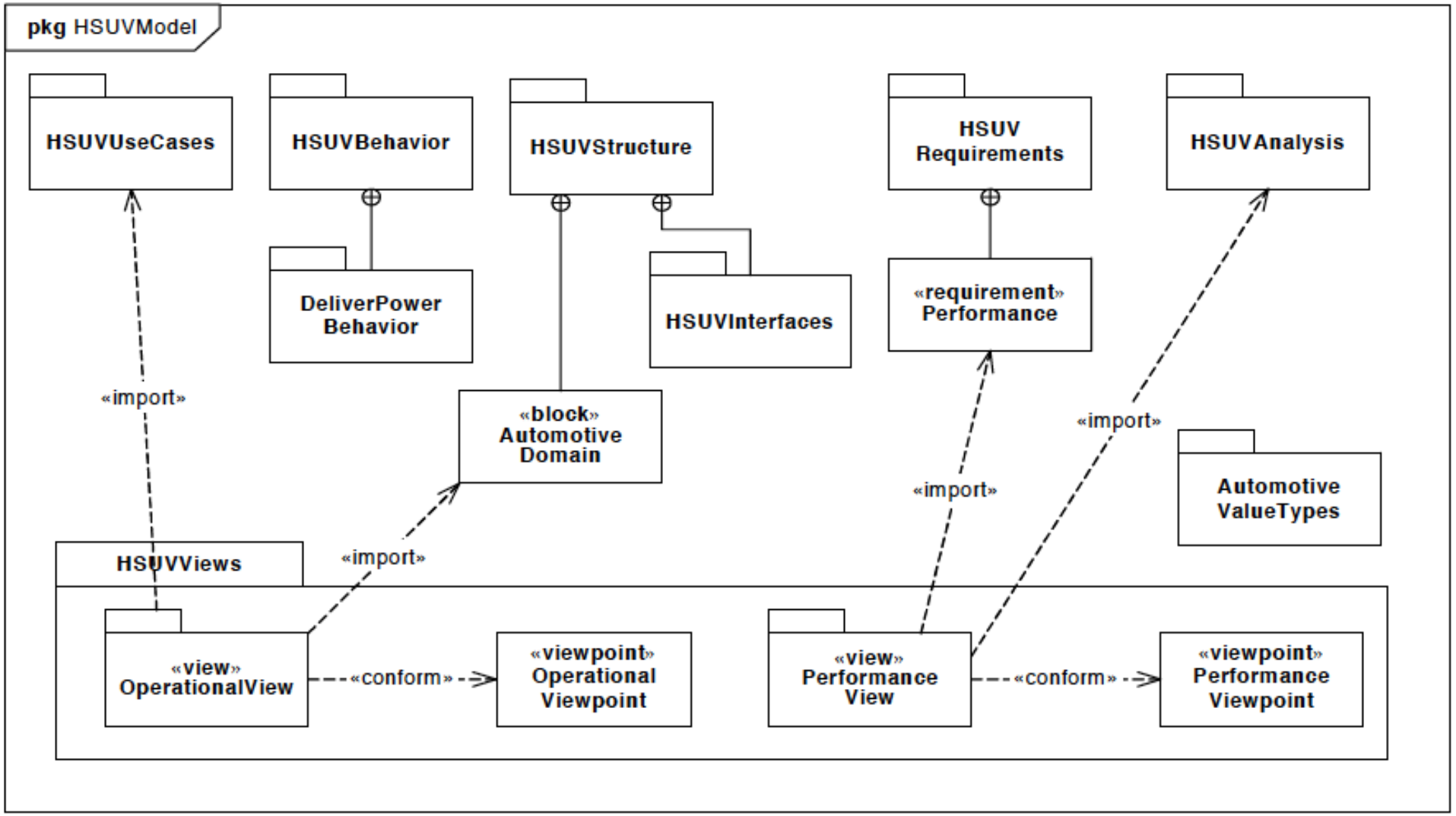
4 Automotive use case “Sample Problem”

specification of units and valueTypes employed in the sample problem



5 Sample Problem

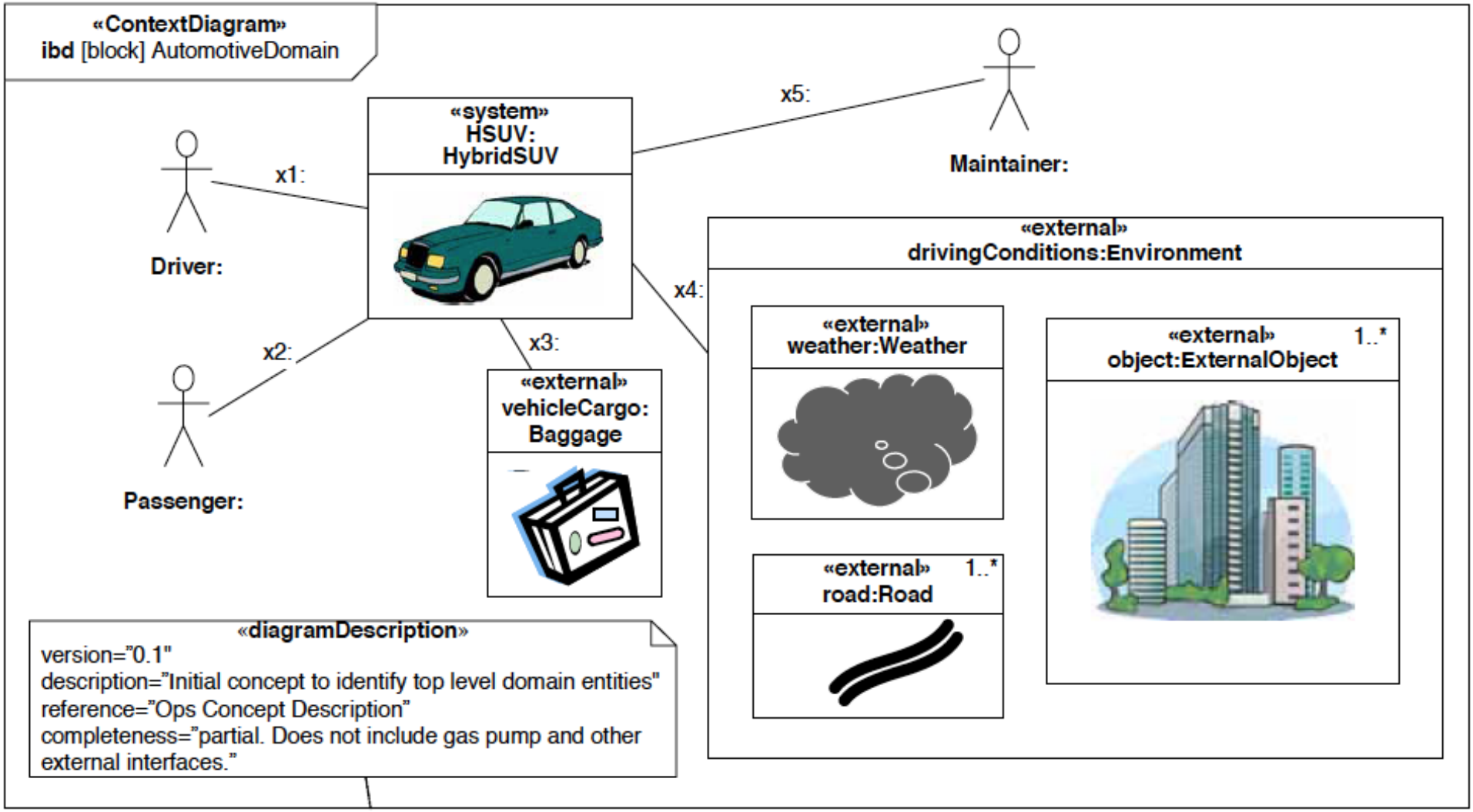
Package Diagram



6 Sample Problem

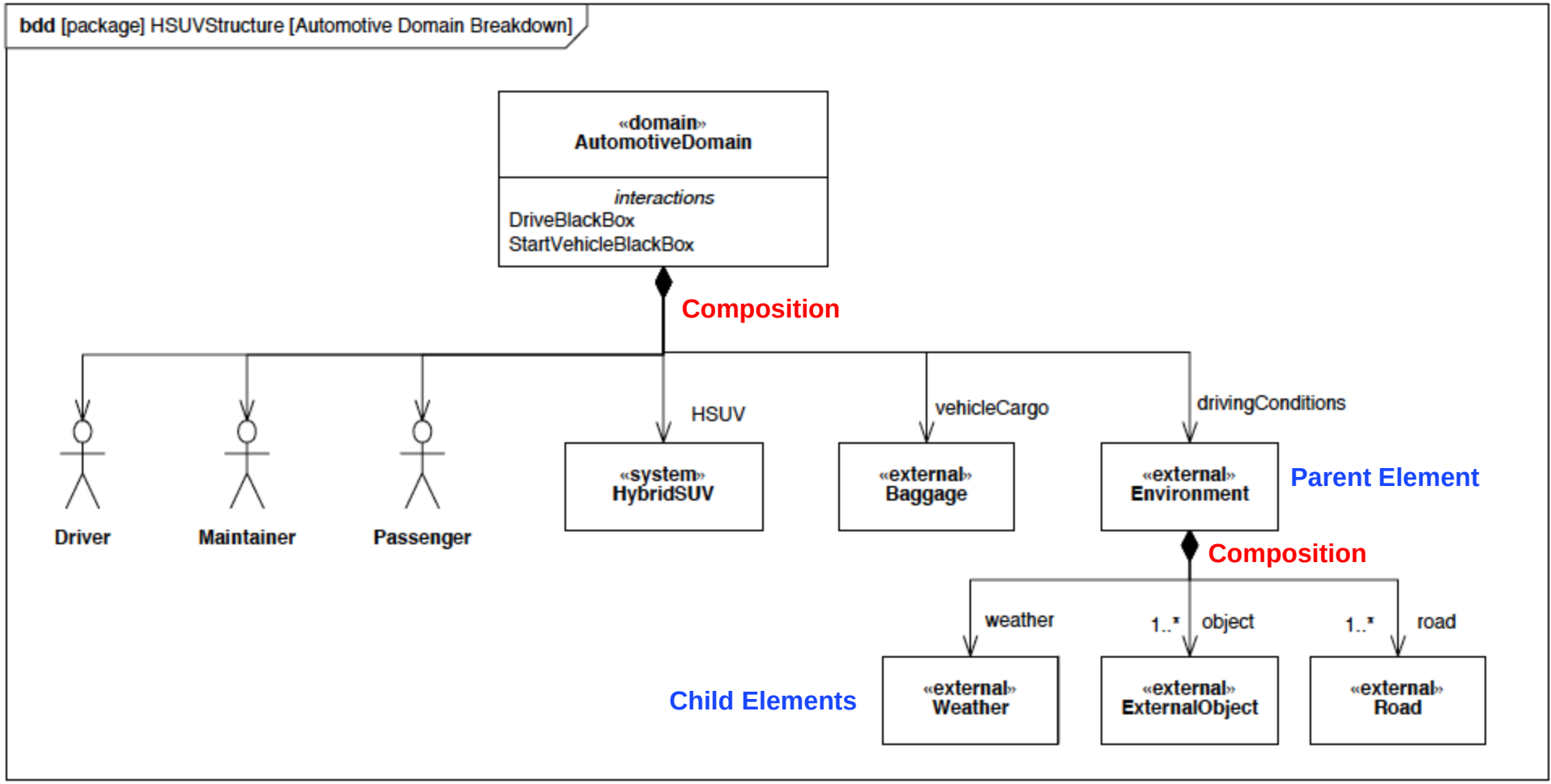
Setting the context

- ... See requirements lecture



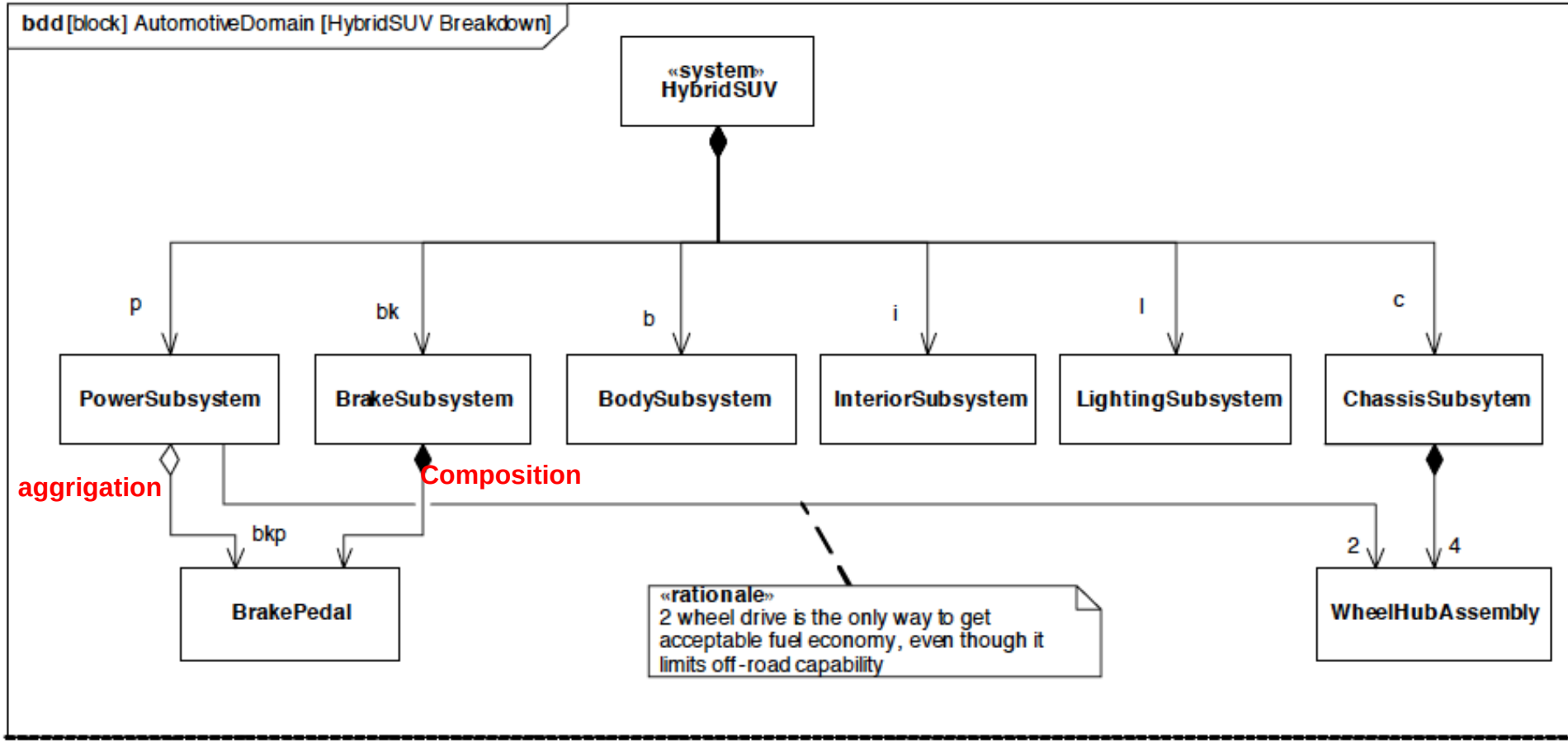
7 Sample Problem

Block Definition Diagram – Automotive Domain



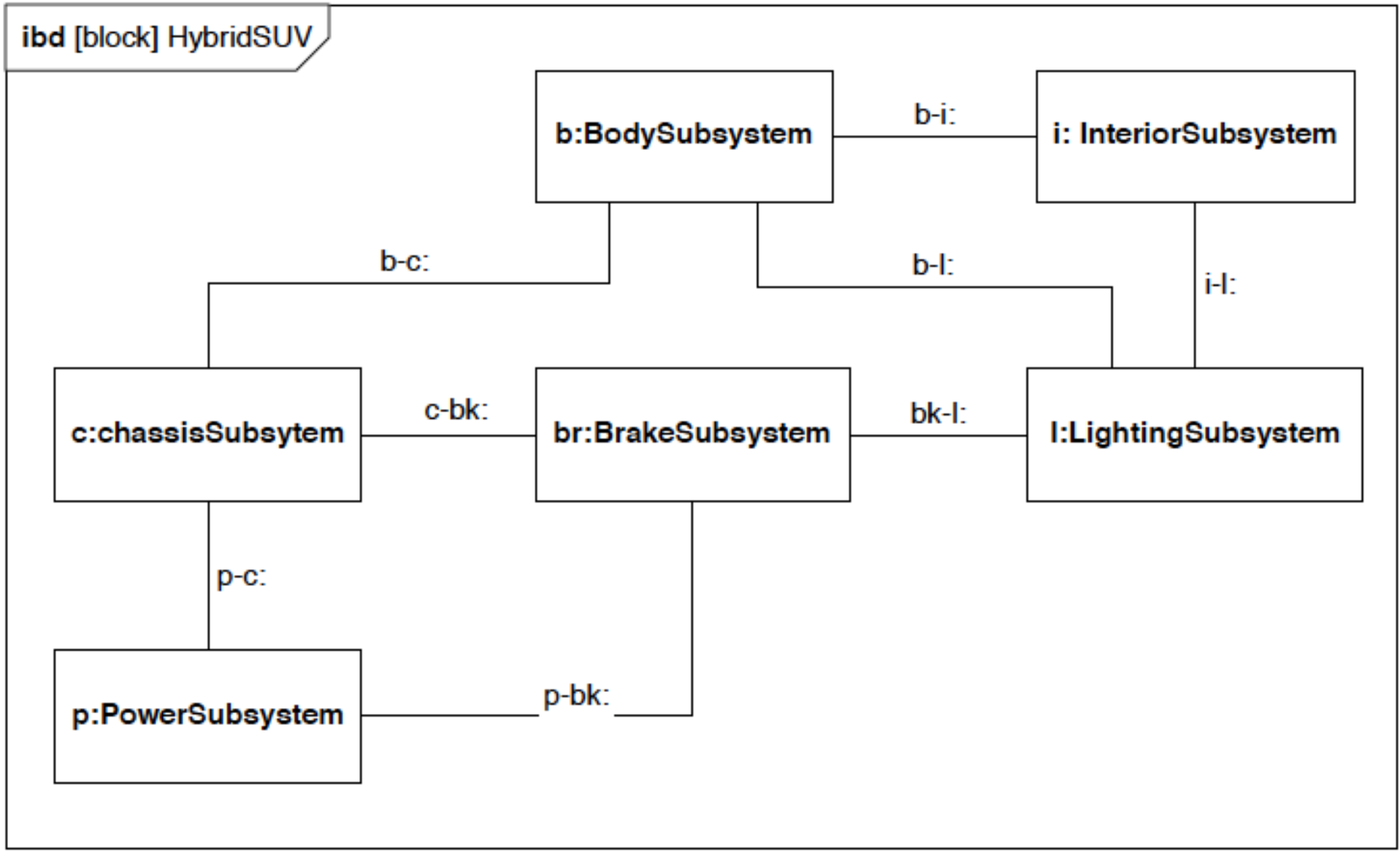
8 Sample Problem

Block Definition Diagram – Hybrid SUV



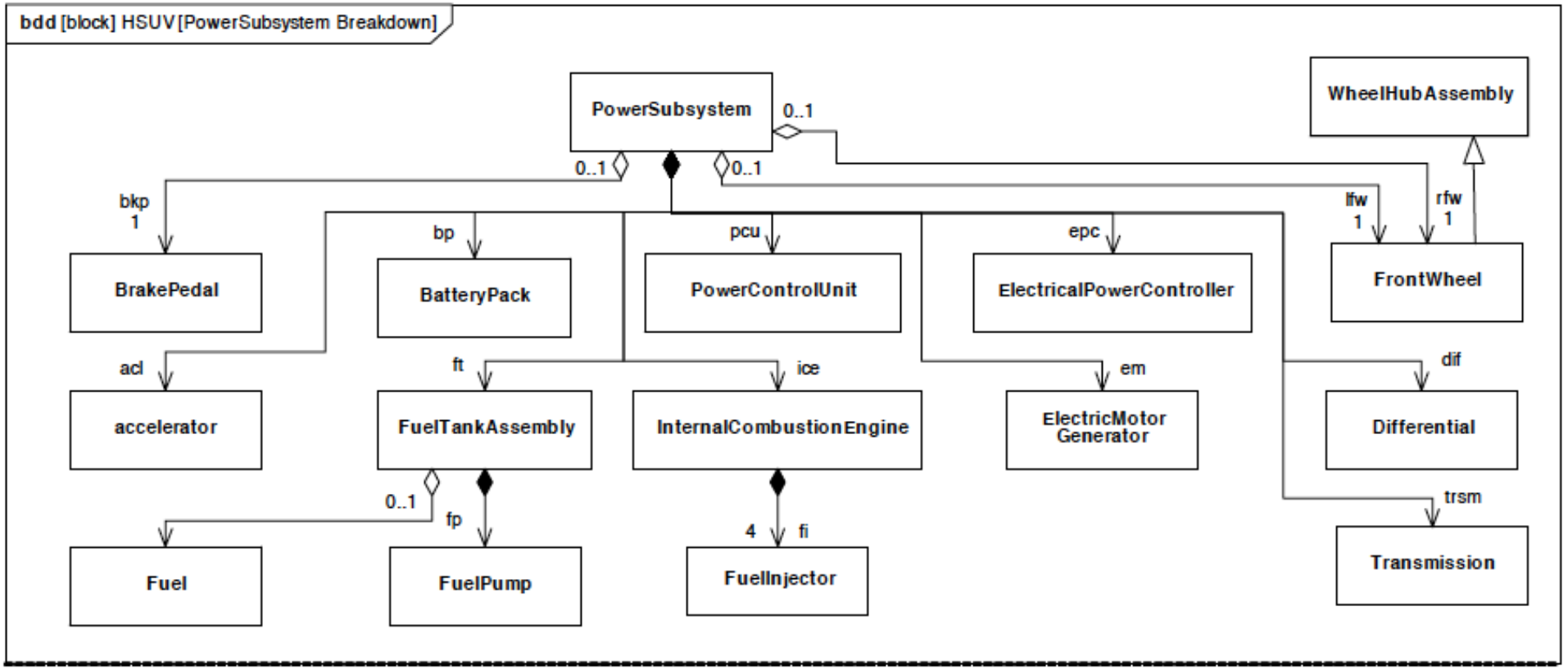
9 Sample Problem

Internal Block Diagram – HSUV



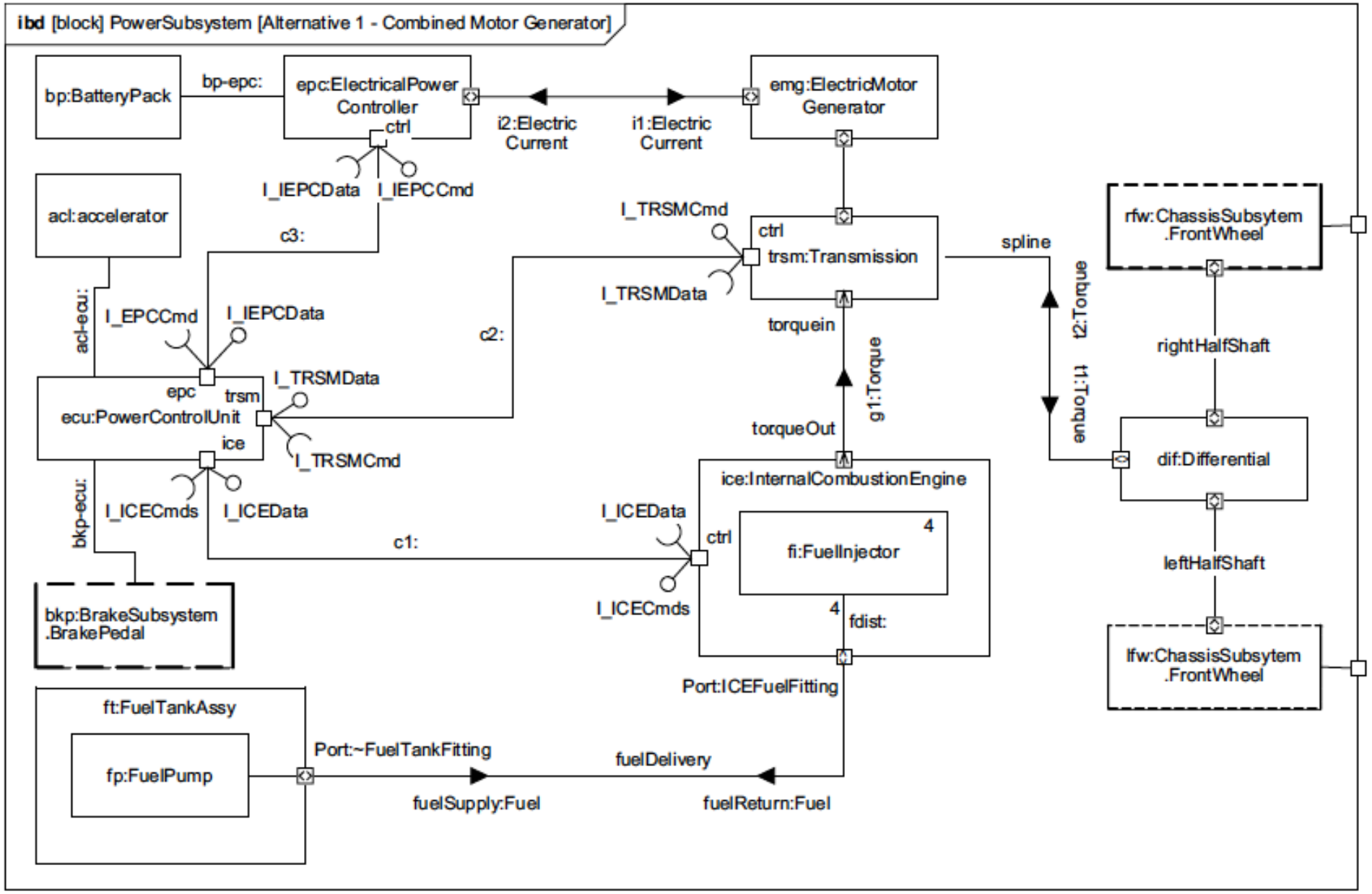
10 Sample Problem

Bdd – Power Subsystem



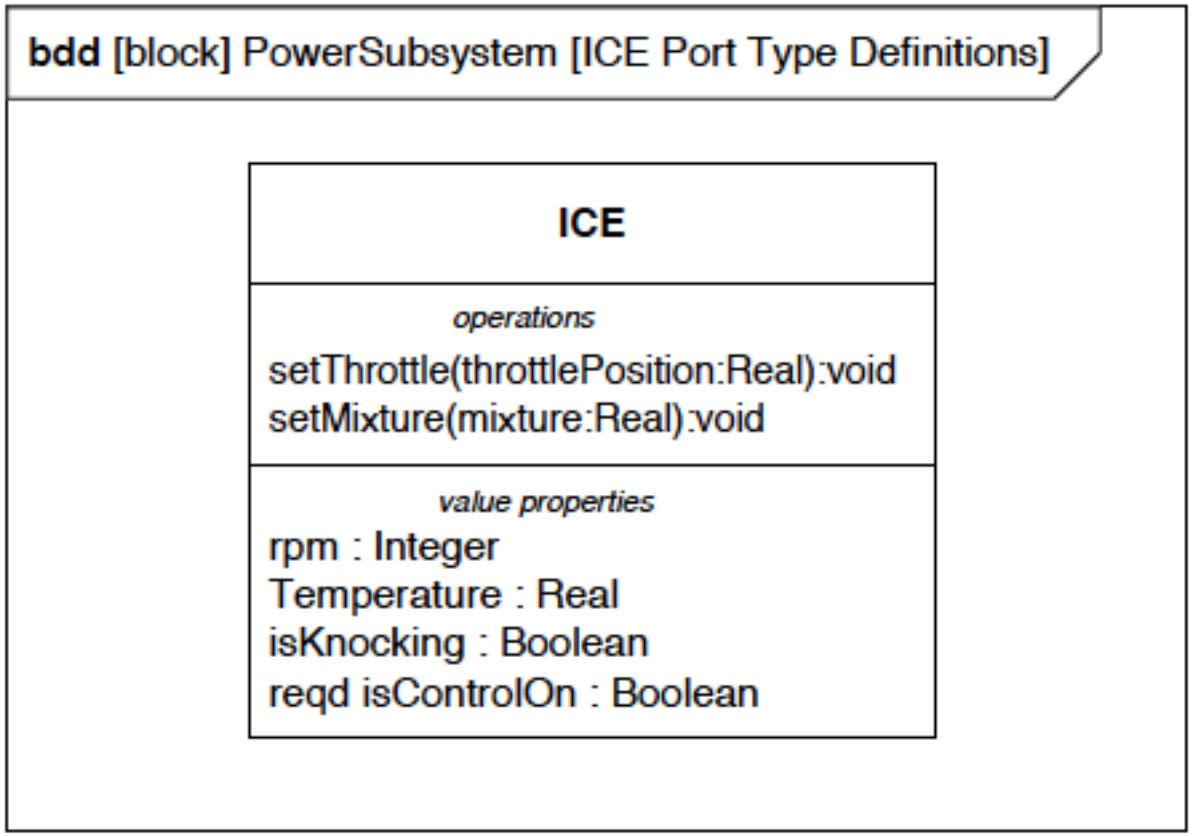
11 Sample Problem

Ibb – Power Subsystem



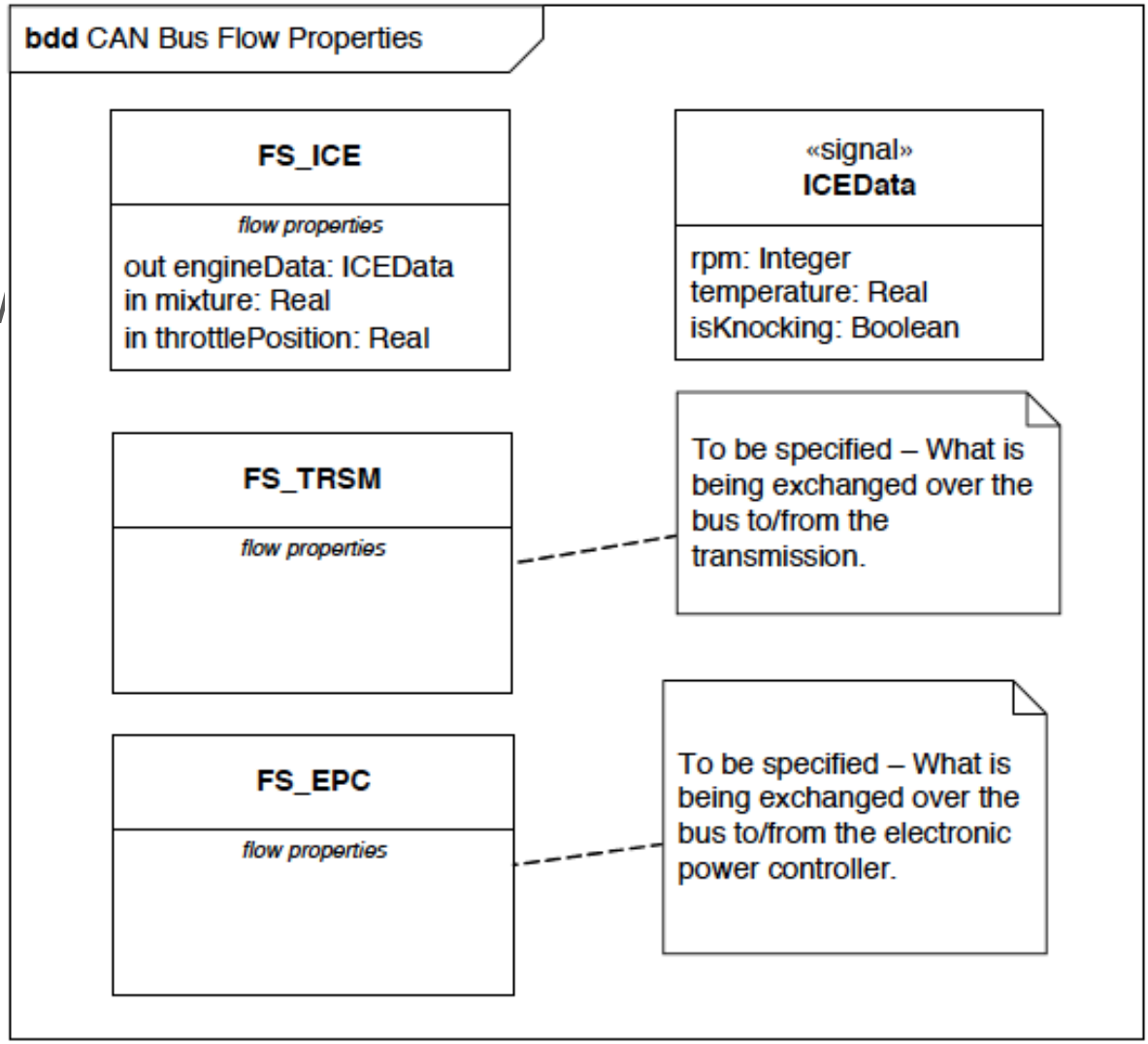
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- Blocks Typing Ports in the Power Subsystem



In Detail: Defining Ports and Flows

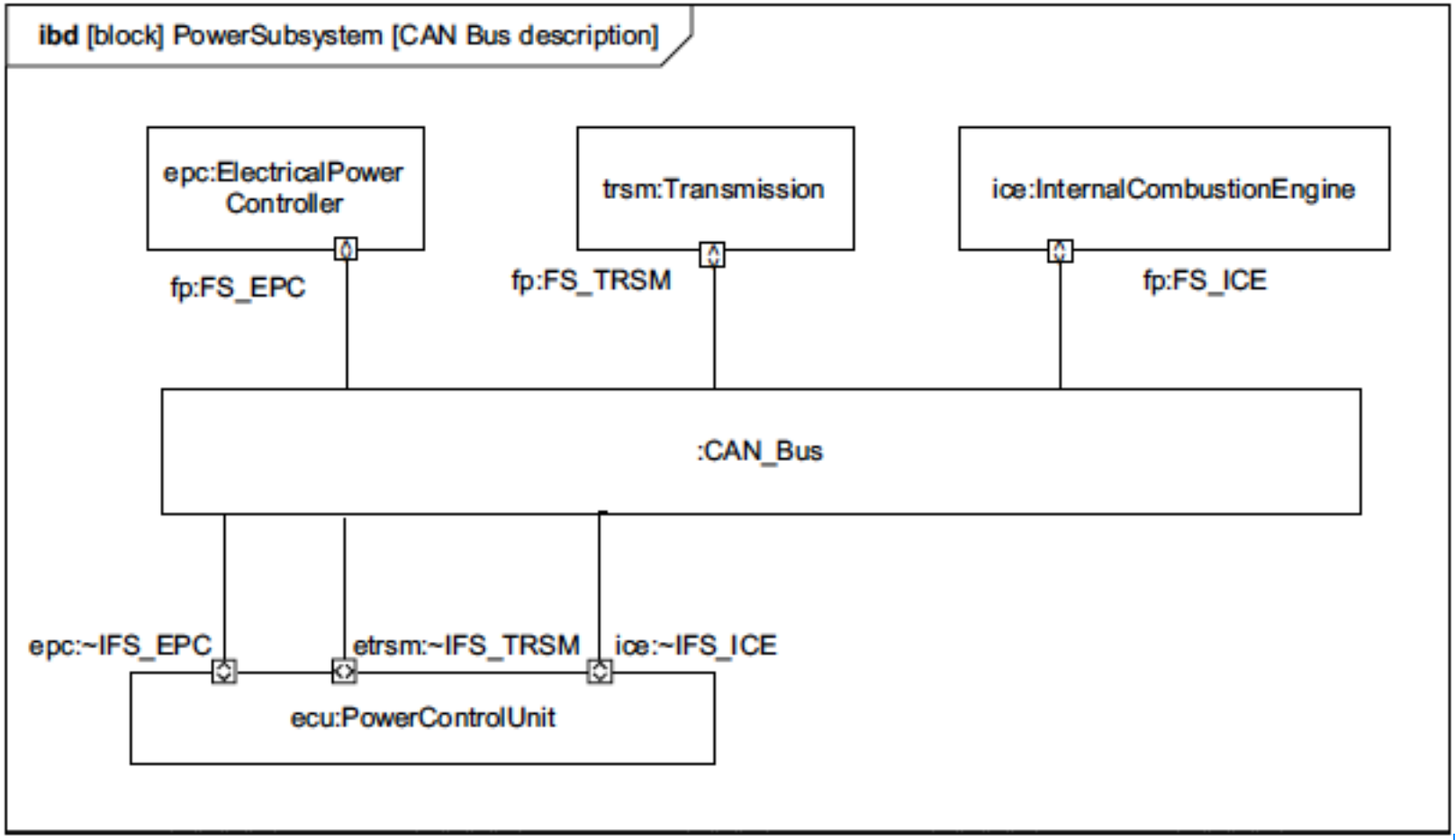
- Use ports with flow properties to model the bus architecture
- Port Types with Flow Properties for the CAN Bus (Block Definition Diagram)



Sample Problem

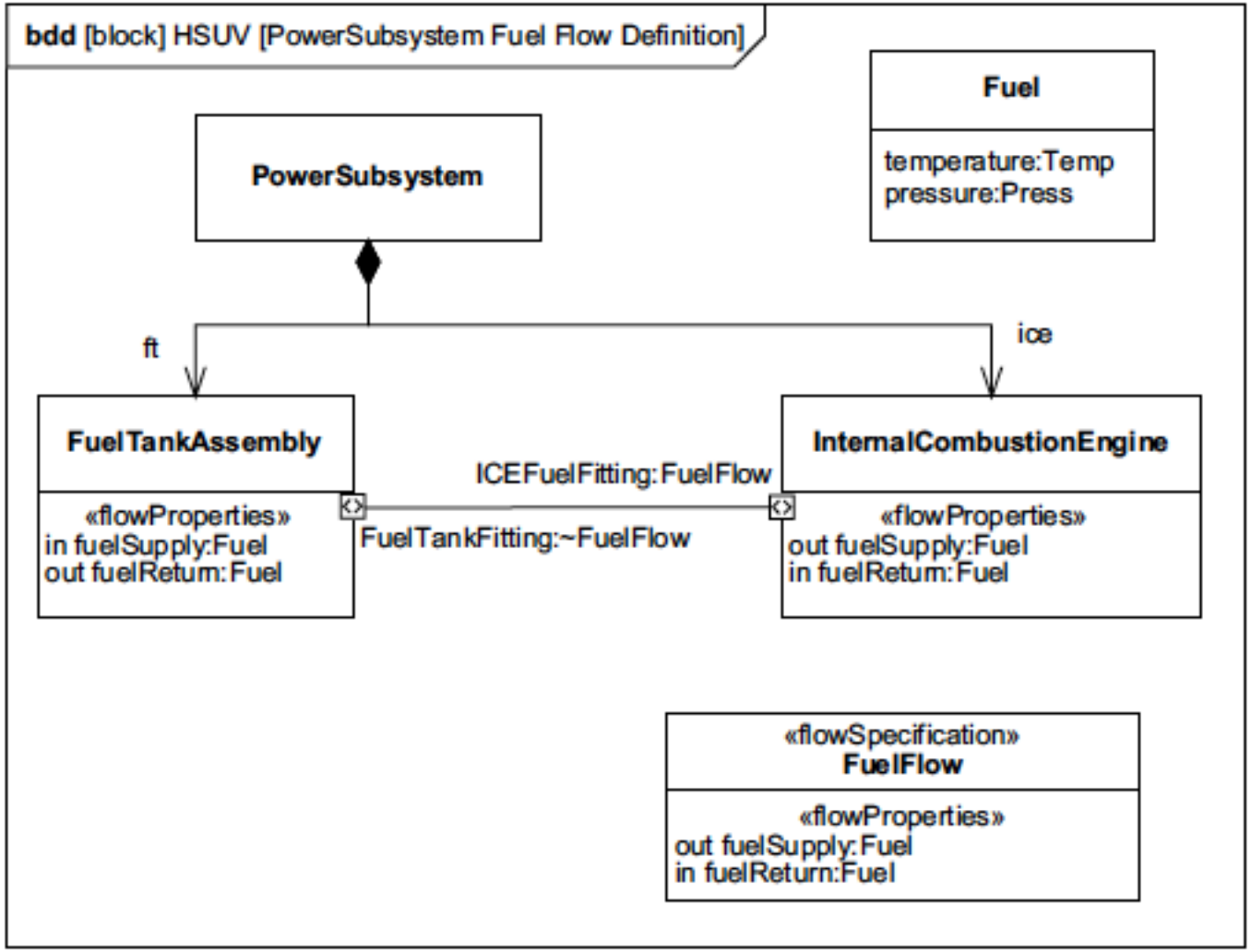
Internal Block Diagram of the CAN-Bus

- refinement of the Controller Area Network (CAN) bus architecture using ports
- Consider explicit structural allocation



Sample Problem

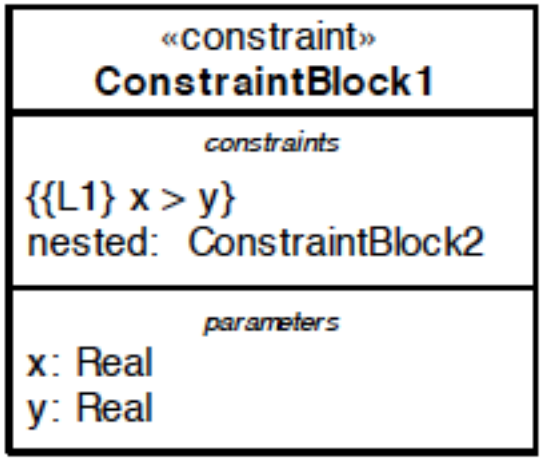
Ports Fuel Tank



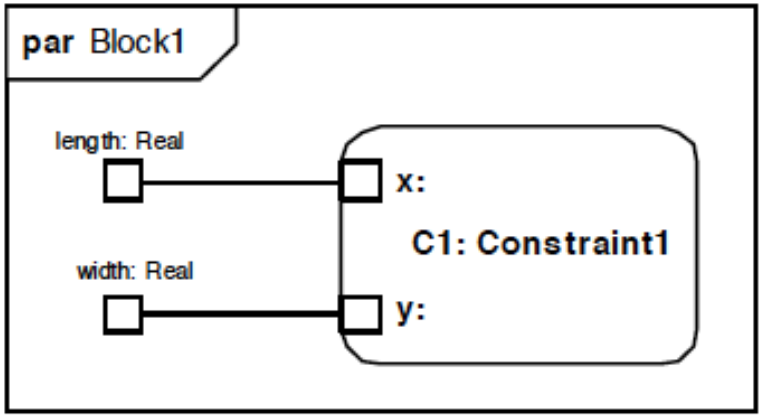
(2) Parametric Constraint Diagram

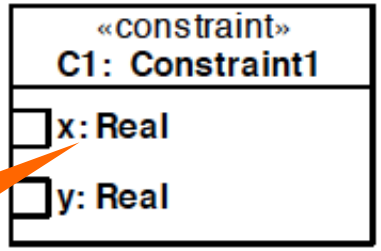
- Can be used to specify a **network of constraints** that represent **mathematical expressions** such as {F=m*a}
  - constrain the physical properties of a system
- Constraints can also be used to **identify critical performance parameters** and their relationships to other parameters
  - can be tracked throughout the system life cycle
- Constraint blocks define **generic forms of constraints** that can be used in multiple contexts
- Usually the constrained properties express **quantitative** characteristics of a system, but parametric models may also be used on **non-quantitative** properties.
- **Time** can be modeled as a property that other properties may be dependent on
  - A time reference can be established by a local or global clock that produces a continuous or discrete time value property.
  - Other values of time can be derived from this clock, by introducing delays and/or skew into the value of time
- **Application**
  - Define dependencies between parameters
  - Define system constraints
  - Parametric constraints are typically used in combination with block diagrams.
  - Can be used to support tradeoff analysis.

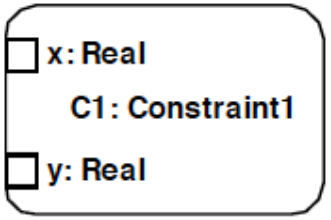
- **ConstraintBlock:** A constraint block is defined by a keyword of «constraint» applied to a block definition. The properties of this block define the parameters of the constraint.



- **Parametric diagrams** include usages of constraint blocks to constrain the properties of another block. The usage of a constraint binds the parameters of the constraint, such as F, m, and a, to specific properties of a block, such as a mass, that provide values for the parameters.





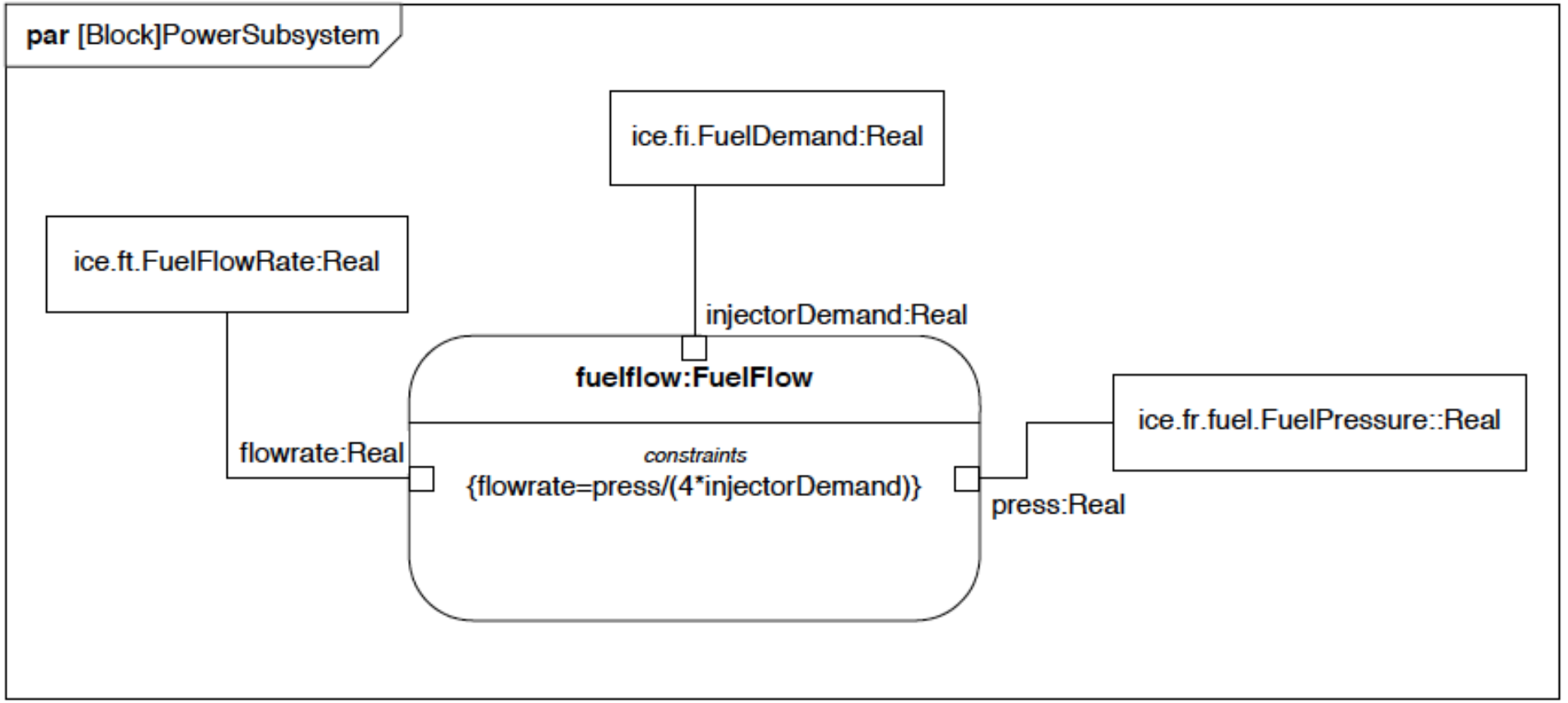


Usage

Example: Sample Problem

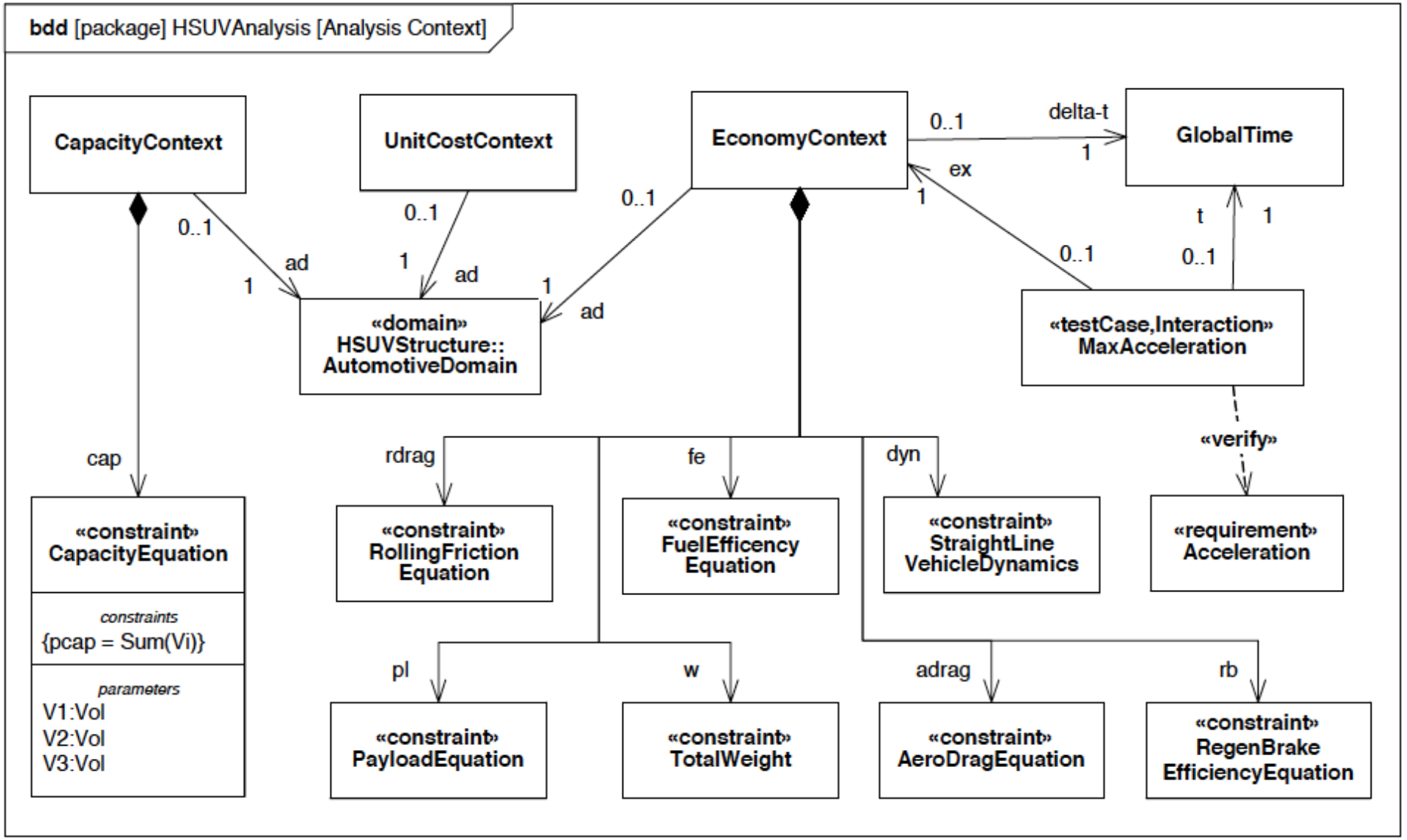
Defining Fuel Flow Constraints

- parametric diagram showing how fuel flowrate is related to FuelDemand and FuelPressure value properties.



Sample Problem

Analysis Context – Constraint Blocks and its relations



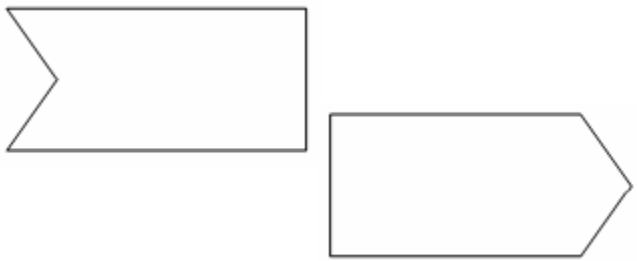
(3) Activity Diagrams

- Activities specify sequential and concurrent behaviors that are connected by **control flows** and **object flows**.
- Activities can be **nested** or **atomic**; in the latter case they are referred to as **actions**.
- Activity diagrams are analogous to **Extended Functional Flow Block Diagrams** (EFFBDs).

- **Application**
  - Can be used to specify the causal/functional behavior of a system.

Basic Actions

- AcceptEventAction/ SendSignalEvent



- Action



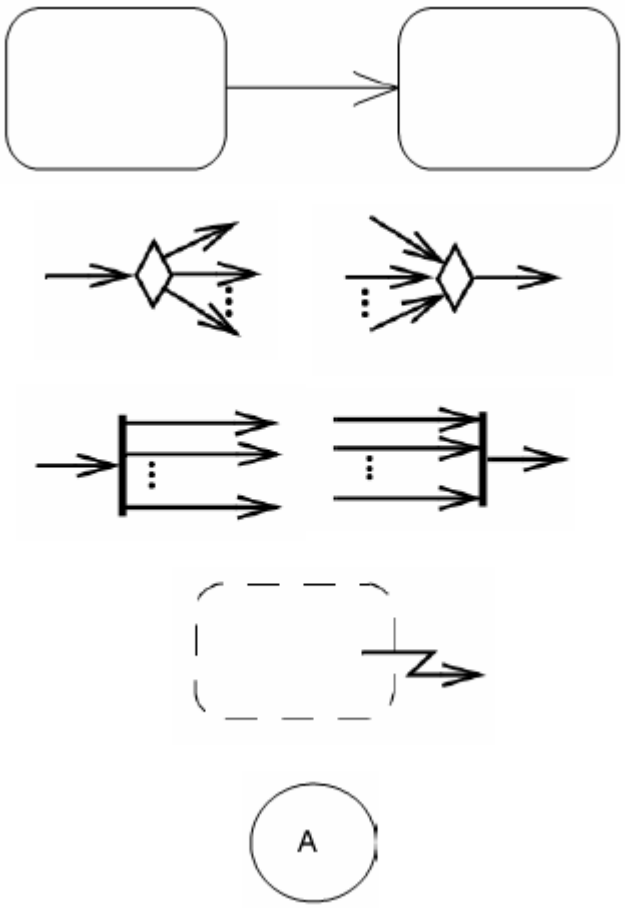
- InitialNode/ActivityFinal



- FlowFinal

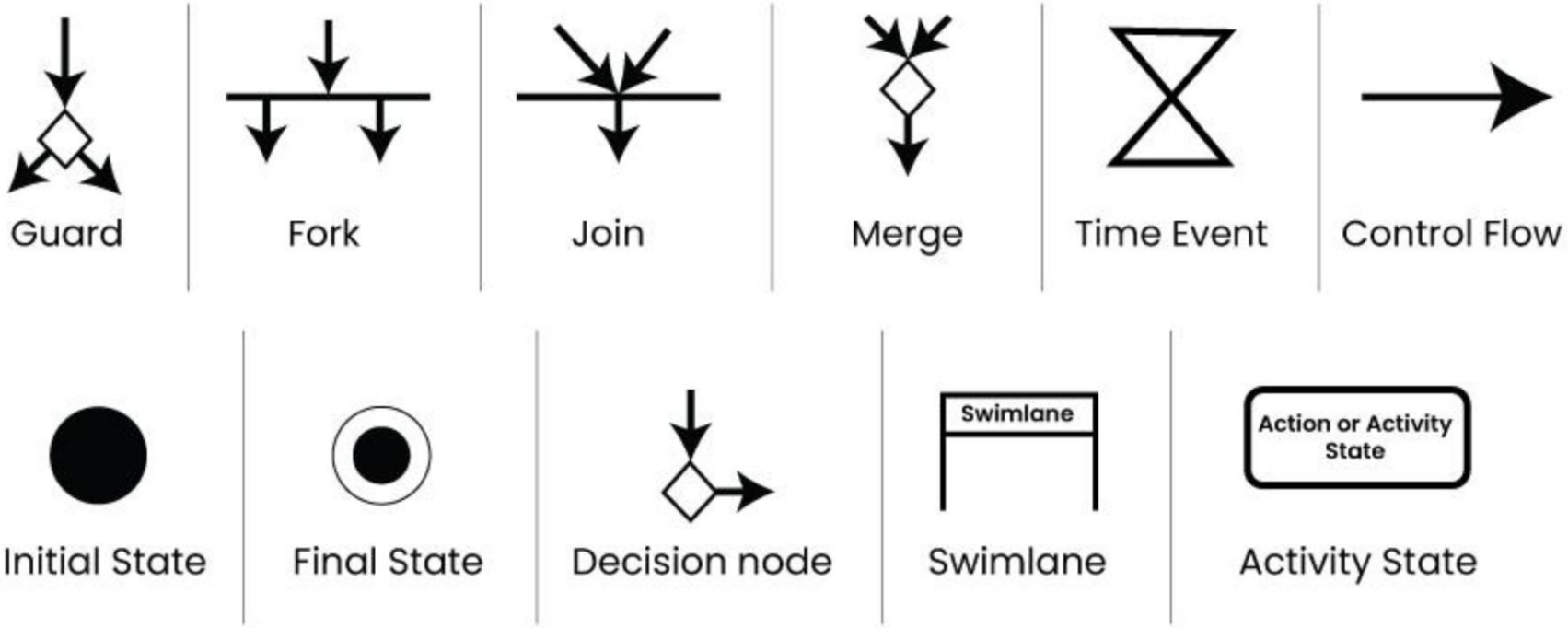


Control Flow



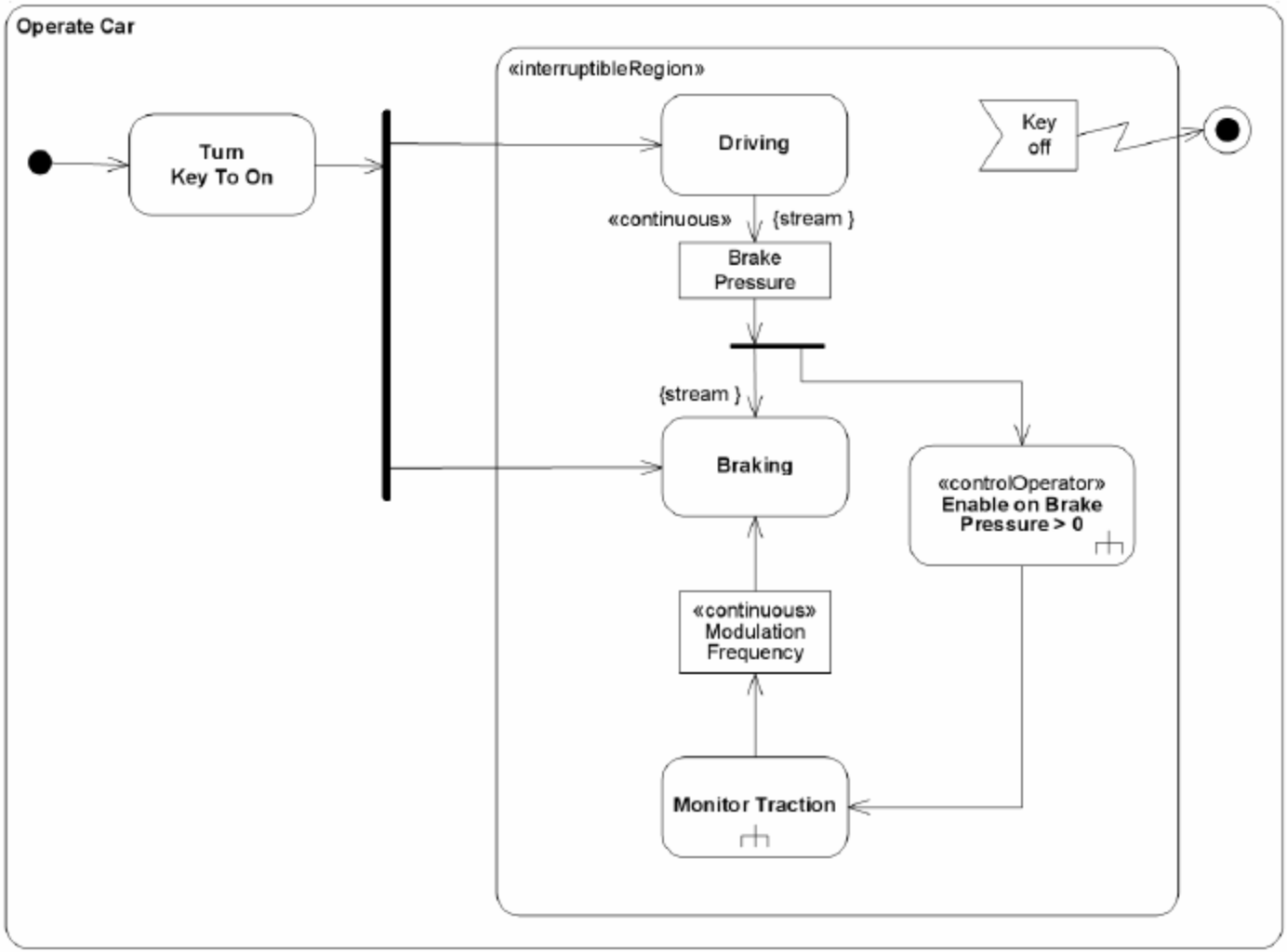
- ControlFlow
- DecisionNode/MergeNode
- ForkNode/JoinNode
- InterruptibleActivityRegion
- Page connectors: used to connect flows across multiple pages.

Activity Diagram Notations



Example

Continuous



Homework

- Consider your Microcontroller project from last semester
  - Specify the parametric constraints and activity diagrams on analysis level
    - Use paper and pen
  - In addition, if you have the possiblity, use the SysML tool papyros
- Readings
  - Tim Weilkiens, “Systems Engineering with SysML/UML” (see: https://learning.oreilly.com/library/view/systems-engineering-with/9780123742742/)